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Enhancing adaptive sparse grid approximations and improving refinement streategies using adjoint-based a posteriori error estimates

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Abstract

In this paper we present an algorithm for adaptive sparse grid approximations of quantities of interest computed from discretized partial differential equations. We use adjoint-based a posteriori error estimates of the interpolation error in the sparse grid to enhance the sparse grid approximation and to drive adaptivity. We show that utilizing these error estimates provides significantly more accurate functional values for random samples of the sparse grid approximation. We also demonstrate that alternative refinement strategies based upon a posteriori error estimates can lead to further increases in accuracy in the approximation over traditional hierarchical surplus based strategies. Throughout this paper we also provide and test a framework for balancing the physical discretization error with the stochastic interpolation error of the enhanced sparse grid approximation.

Keywords: Uncertainty quantification, a posteriori error estimation, sparse grids, stochastic collocation, adaptivity

1. Introduction

Partial Differential Equations (PDE) are used to simulate a wide range of phenomenon and are often used to inform design decisions and to estimate risk in systems with large human and/or financial impact but with

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limited capacity for experimentation. Given the importance of these applications the ability to accurately quantify uncertainty in model predictions is essential.

Most uncertainty quantification (UQ) studies focus on estimating parametric uncertainty. In such analyses, the uncertainty in the input data, such as model coefficients, forcing terms etc, is usually represented through a finite number of random variables with a known probability distribution. The goal of the study is then to compute the effect of the varying input data on the system response, and in many cases, to calculate the statistics of the response.

The accuracy to which uncertainty can be quantified is limited by the computational resources available to resolve these governing equations. Many applications require vast amounts of computational effort and thus the number of model evaluations that can be used to interrogate the uncertainty in the system behaviour is limited. Consequently a significant portion of methods developed for uncertainty quantification (UQ) in recent years have focused on constructing surrogates of expensive simulation models using only a limited number of model evaluations.

The most widely adopted approximation methods are based on generalized polynomial chaos (PC) expansions [12, 21], sparse grid interpolation [16, 17] and Gaussian Process (GP) models. The performance of these methods is problem dependent and in practice it is difficult to estimate the accuracy of the approximation constructed. Cross-validation is one means of estimating the accuracy of the approximation, however the accuracy of the cross-validation prediction of the error is limited. Moreover, cross validation is not possible for approximation methods which require structured model samples, such as sparse grid interpolation and many forms of pseudo-spectral projection.

Recently, a posteriori error estimation has arisen as a promising approach to estimate the error in approximate input-output relationships. Adjoint-based a posteriori error estimation was originally developed to estimate error in numerical approximations of deterministic PDEs [13, 8, 1, 19], but recent modifications allow similar ideas to be used to estimate error in approximations of quantities of interest from PDEs with uncertain parameters. This relatively new approach, introduced in [5] and further analyzed in [4, 6], is based on goal-oriented adjoint-based error estimates and is used to predict error in samples of a response surface approximation of a specific quantity of interest. Similar to standard adjoint-based error estimation procedures, this new approach includes the physical discretization error if the adjoint problem is approximated in a higher-order discretization space. However,

the error estimate from this new approach also contains an approximation of the error in the stochastic discretization due to the evaluation of the response surface model rather than the PDE. In [4, 6], it was shown that, for spectral and pseudo-spectral Galerkin approximations, this estimate of the stochastic interpolation error is higher-order even if a low order approximation of the adjoint is used for the stochastic approximation.

In this paper we utilize sparse grid interpolation to approximate model responses. Sparse grids can be built using local or global basis functions and have well established and effective adaptivity procedures which can be leveraged in conjunction with good error estimates to concentrate computational effort to resolving important dimensions and/or regions of the random parameter space. Unlike regression based PCE or Gaussian Process models, sparse grid interpolation also has the flexibility to easily deal with vector valued responses with large numbers of entries and can be used regardless of the computational budget of the UQ analysis. For example sparse grids can be used to approximate a model response using tens to millions of model runs, whereas the aforementioned alternatives have limits in the low thousands imposed by the need to solve large linear systems.

Our goal in this paper is to utilize adjoint-based a posteriori error estimates to efficiently compute pointwise approximations of specific quantities of interest, usually computed from PDE solutions, using adaptive sparse grid approximations. Specifically, we aim to

- Show that the enhancement results in [4] also extend to adaptive sparse grid approximations.
- Present new adaptivity strategies for sparse grids based on a posteriori error estimates.
- Demonstrate that, for a given computational budget, better accuracy can be obtained using the proposed approach.
- Provide a stopping criteria, based on the physical discretization error, to avoid over-resolving the adaptive sparse grid.

The remainder of this paper is organized as follows. Section 2 introduces a general model problem. In Section 3 we recall the standard adjoint-based posteriori error analysis for deterministic PDEs. Section 4 presents the standard adaptive sparse grid algorithms. In Section 5 we derive an a posteriori error estimate for samples of a sparse grid surrogate. In Section 6, we present the new adaptive strategies using the error estimates. Section 7 introduces the sparse grid approximation of the error estimate and our stopping criteria

based on an estimate the physical discretization error. Numerical results are presented in Section 8 and our conclusions are presented in Section 9.

2. General nonlinear problem and notation

We consider the following system of partial differential equations,

$$\frac{\partial \mathbf{z}}{\partial t} + \mathbf{A}(\boldsymbol{\xi}; \mathbf{z}) = \mathbf{0}, \tag{2.1}$$

defined on $\Omega \times (0, T]$ where $\Omega \subset \mathbb{R}^s$, $s = 2, 3$, a polygonal (polyhedral) and bounded domain with boundary $\partial\Omega$. The random parameter $\boldsymbol{\xi}$ takes values in $\Gamma_{\boldsymbol{\xi}} \subset \mathbb{R}^d$ and reflects uncertainty in model and source parameters. The solution operator's dependency on $\boldsymbol{\xi}$ implies that $\mathbf{z} := \mathbf{z}(x, t, \boldsymbol{\xi})$ is also uncertain and may be modeled as a random process for which we will construct a surrogate approximation. We assume that $\mathbf{A}$ is convex and has smooth second derivatives. Specific examples of $\mathbf{A}$ and $\mathbf{z}$ will be given in subsequent sections. We assume that sufficient initial and boundary conditions are provided so that (2.1) is well-posed in the sense that there exists a solution for a. e. $\boldsymbol{\xi} \in \Gamma_{\boldsymbol{\xi}}$.

We use $(\cdot, \cdot)_{\mathcal{D}}$ to denote the inner product of $\boldsymbol{L}^2(\mathcal{D})$ and if the domain of integration is clear from the context, we suppress the index $\mathcal{D}$. We let $\boldsymbol{V}$ be a Sobolev space where for any non-negative integer m we recall

$$H^m(\Omega) = \{v \in L^2(\Omega)\,;\, \partial^k v \in L^2(\Omega) \,\forall\, |k| \le m\},$$

equipped with the following seminorm and norm

$$|v|_{H^m(\Omega)} = \left[\sum_{|k|=m} \int_\Omega |\partial^k v|^2 \, dx\right]^{1/2}, \quad \|v\|_{H^m(\Omega)} = \left[\sum_{0 \le |k| \le m} |v|^2_{H^k(\Omega)}\right]^{1/2}.$$

We frequently use $\boldsymbol{H}^m(\Omega)$ to denote the obvious generalization to vector valued functions.

We assume that (2.1) has an equivalent variational formulation seeking, for any $\boldsymbol{\xi} \in \Gamma_{\boldsymbol{\xi}}$, $\mathbf{z} \in \boldsymbol{V} \times (0, T]$ such that,

$$\int_0^T \left[\left(\frac{\partial \mathbf{z}}{\partial t}, \mathbf{w}\right) + \boldsymbol{a}(\boldsymbol{\xi}; \mathbf{z}, \mathbf{w})\right] \, dt = \mathbf{0}, \quad \forall \mathbf{w} \in \boldsymbol{V} \times (0, T]. \tag{2.2}$$

Let $\mathcal{T}_h$ be a conforming partition of Ω, composed of N_T closed convex volumes of maximum diameter h. An element of the partition $\mathcal{T}_h$ will be

denoted by T_i where h_i stands for the diameter of T_i for $i = 1, 2, \ldots, N_T$. We assume that the mesh is regular in the sense of Ciarlet [7]. In this paper we take $\mathcal{T}_h$ to be a conforming finite element mesh consisting of simplices or parallelopipeds. Let the space of continuous, piecewise functions of degree q over the spatial domain be defined by

$$\boldsymbol{V}_h^{(q)} = \left\{ \mathbf{v} \in C(S) \cap \boldsymbol{H}^1(\Omega) : \forall E \in \mathcal{T}_h, \mathbf{v}|_E \in \mathbb{P}^q(E) \right\}.$$

The semi-discrete variational formulation seeks, for any $\boldsymbol{\xi} \in \Gamma_{\boldsymbol{\xi}}$, $\mathbf{z}_h \in \boldsymbol{V}_h^{(q)} \times (0, T]$ such that,

$$\int_0^T \left[\left(\frac{\partial \mathbf{z}_h}{\partial t}, \mathbf{w} \right) + \boldsymbol{a}(\boldsymbol{\xi}; \mathbf{z}_h, \mathbf{w}) \right] \, dt = \mathbf{0}, \quad \forall \mathbf{w} \in \boldsymbol{V}_h^{(q)} \times (0, T]. \tag{2.3}$$

A fully discrete scheme for any $\boldsymbol{\xi} \in \Gamma_{\boldsymbol{\xi}}$ can be obtained by letting $I_n = (t_{n-1}, t_n)$ and time steps $k_n = t_n - t_{n-1}$ denote the discretization of $[0, T]$ as $0 = t_0 < t_1 < \cdots < t_N = T$. In this paper, we do not focus on any particular time integration scheme. We do assume that the fully discrete approximation is given as a polynomial in time, e.g.,

$$\mathbf{z}_h \in \boldsymbol{V}_h^{(q)} \times \mathcal{W}^{(r)},$$

where

$$\mathcal{W}^{(r)} = \left\{ w \in C([0, T]) : w \in \mathbb{P}^r(I_n), \forall I_n \right\},$$

denotes the space of continuous piecewise polynomial functions of degree r. To make the notation less cumbersome, we generally use $\mathbf{z}_h$ to denote the fully discrete solution unless otherwise noted. The choice of continuous polynomials in time is merely for convenience. Discontinuous polynomials may also be used and require a straightforward modification to the a posteriori error analysis involving jump terms at each time node [9, 8].

3. Adjoint-based a posteriori error estimation

Predictive simulations often aim to accurately estimate a relatively small number of quantities of interest. The goal in adjoint-based error analysis is to relate the error ($\mathbf{e} = \mathbf{z} - \mathbf{z}_h$) in a quantity of interest to a computable weighted residual. We are usually interested in estimating the error in a numerical approximation computed using a discretization of a variational formulation, but it is often intuitive to start the discussion with strong form

adjoint operators. We assume that we are interested in estimating the error in a linear functional of the solution,

$$J(\boldsymbol{\xi}) = (\boldsymbol{\psi}_T(x), \mathbf{z}(x, T, \boldsymbol{\xi})) + \int_0^T (\boldsymbol{\psi}, \mathbf{z}(x, t, \boldsymbol{\xi}))\ dt, \tag{3.1}$$

where $\boldsymbol{\psi}_T(x)$ is used to compute a linear functional at $t = T$, and $\boldsymbol{\psi} = \boldsymbol{\psi}(x, t)$ is used to compute time-averages. Typically, either $\boldsymbol{\psi}_T$ or $\boldsymbol{\psi}$ are zero, but this is not required. Nonlinear functionals and linear functionals at $t \neq T$ can also be used, but this is omitted here for the sake of simplicity.

For a given $\boldsymbol{\xi} \in \Gamma_{\boldsymbol{\xi}}$, the linear adjoint operator in strong form is defined via the duality relation

$$\int_0^T \left[\left(\frac{\partial \mathbf{z}}{\partial t}, \mathbf{v} \right) + (\boldsymbol{\mathcal{L}}(\boldsymbol{\xi})\mathbf{v}, \mathbf{w}) \right]\ dt = \int_0^T \left[\left(-\frac{\partial \mathbf{v}}{\partial t}, \mathbf{z} \right) + (\mathbf{v}, \boldsymbol{\mathcal{L}}^*(\boldsymbol{\xi})\mathbf{w}) \right]\ dt \tag{3.2}$$

where $\boldsymbol{\mathcal{L}}(\boldsymbol{\xi})$ is a given linear operator. For a general nonlinear PDE one approach to define the linear operator $\boldsymbol{\mathcal{L}}(\boldsymbol{\xi})$ is to assume $\mathbf{A}(\boldsymbol{\xi}; \cdot)$ is convex and use the Integral Mean Value Theorem yielding

$$\mathbf{A}'(\boldsymbol{\xi}, \overline{\boldsymbol{z}}; \mathbf{e}) = \mathbf{A}(\boldsymbol{\xi}; \mathbf{z}) - \mathbf{A}(\boldsymbol{\xi}; \mathbf{z}_h)$$

where $\overline{\boldsymbol{z}}$ lies on the line connecting $\mathbf{z}$ and $\mathbf{z}_h$, and $\mathbf{e} = \mathbf{z} - \mathbf{z}_h$. In practice, $\overline{\boldsymbol{z}}$ is unknown so we linearize around $\mathbf{z}_h$ giving,

$$\boldsymbol{\mathcal{L}}(\boldsymbol{\xi})\mathbf{e} = \mathbf{A}'(\boldsymbol{\xi}, \mathbf{z}_h; \mathbf{e}) = \mathbf{A}(\boldsymbol{\xi}; \mathbf{z}) - \mathbf{A}(\boldsymbol{\xi}; \mathbf{z}_h) + \boldsymbol{R}(\boldsymbol{\xi}; \mathbf{e}, \mathbf{z}_h, \overline{\boldsymbol{z}}),$$

where $\boldsymbol{R}(\boldsymbol{\xi}; \mathbf{e}, \mathbf{z}_h, \overline{\boldsymbol{z}})$ represents the remainder. Since $\mathbf{z}_h - \overline{\boldsymbol{z}} \approx \mathbf{e}$, it is common to assume that the remainder is a higher order perturbation term and can be neglected [2, 1, 9]. Notice that the operator $\boldsymbol{\mathcal{L}}(\boldsymbol{\xi})$ is often the same linear operator used in computing the step in Newton's method. This fact is often exploited to ease construction of the discrete adjoint operator.

The error in a linear functional can be represented using the definition of the adjoint:

$$\begin{aligned} J(\boldsymbol{\xi}; \mathbf{z}) - J(\boldsymbol{\xi}; \mathbf{z}_h) = (\boldsymbol{\phi}(x, 0, \boldsymbol{\xi}), \mathbf{z}(x, 0, \boldsymbol{\xi}) - \mathbf{z}_h(x, 0, \boldsymbol{\xi})) - \\ \int_0^T \left(\left(\frac{\partial \mathbf{z}_h}{\partial t}, \boldsymbol{\phi} \right) + \boldsymbol{a}(\boldsymbol{\xi}; \mathbf{z}_h, \boldsymbol{\phi}) \right)\ dt, + \int_0^T \boldsymbol{R}(\boldsymbol{\xi}; \mathbf{e}, \mathbf{z}_h, \overline{\boldsymbol{z}}, \boldsymbol{\phi})\ dt, \end{aligned} \tag{3.3}$$

where $\boldsymbol{\phi} := \boldsymbol{\phi}(x, t, \boldsymbol{\xi})$ is defined by the adjoint problem

$$-\frac{\partial \boldsymbol{\phi}}{\partial t} + \boldsymbol{\mathcal{L}}^*(\boldsymbol{\xi})\boldsymbol{\phi} = \boldsymbol{\psi}(x, t), \tag{3.4}$$

$$\phi(x, T) = \boldsymbol{\psi}_T(x). \tag{3.5}$$

If the adjoint solution, $\boldsymbol{\phi}$, is given, then the error representation in Eq. (3.3) is easily evaluated if we neglect the higher order remainder term, see [2, 1, 9, 10] for a complete discussion of this remainder term. However, the solution to the adjoint problem Eq. (3.4) is usually not given explicitly and we must approximate the solution using an appropriate discretization. In this paper, we approximate the adjoint solution using a finite element method. Since we are also interested in estimating the physical discretization error, we use a higher-order approximation in the spatial domain. In general, the definition of the adjoint problem also requires appropriate boundary conditions and the error representation (3.3) also includes boundary terms to account for errors made in approximating the boundary conditions on the forward problem. We omit these terms in this paper for the sake of simplicity and refer the interested reader to the standard references, e.g. [13, 8, 1, 19], for a complete discussion of these boundary terms.

4. Sparse grid approximations

Sparse grid stochastic collocation has been shown to provide efficient and accurate approximation of stochastic quantities [16, 17, 18]. Here we adopt this technique to construct approximations of functionals of the discretized solution, $J_h(\boldsymbol{\xi})$. We typically do not know the closed form of $J_h(\boldsymbol{\xi})$ and only require that we can evaluate $J_h(\boldsymbol{\xi})$ at arbitrary points in $\Gamma_{\boldsymbol{\xi}}$ by evaluating the discretized model. For simplicity we will restrict attention to consider stochastic collocation problems characterized by variables $\boldsymbol{\xi}$ with finite support normalized to fit in the domain $\Gamma_{\boldsymbol{\xi}} = [0,1]^d$. However the technique proposed here can be applied to semi or unbounded random variables using the methodology outlined in [15].

Sparse grids [3] approximate $J_h(\boldsymbol{\xi})$ via a weighted linear combination of basis functions

$$I_n[J_h(\boldsymbol{\xi})] := J_{h,n}(\boldsymbol{\xi}) = \sum_{k=1}^{n} v_k \, \Psi_k(\boldsymbol{\xi}) \tag{4.1}$$

The approximation is constructed on a set of anisotropic grids $\Xi_{\mathbf{l}}$ on the domain $\Gamma_{\boldsymbol{\xi}}$ where $\mathbf{l} = (l_1, \ldots, l_d) \in \mathbb{N}^d$ is a multi-index denoting the level of refinement of the grid in each dimension. These rectangular grids are Cartesian product of nested one-dimensional grid points $\Xi_l = \{\xi_{l,i} : i < 0 \leq i \leq m_l\}$

$$\Xi_{\mathbf{l}} = \Xi_{l_1} \times \cdots \Xi_{l_d}$$

Typically when approximating $J_h(\boldsymbol{\xi})$ with a smooth dependence on $\boldsymbol{\xi}$, Ξ_l are chosen to be the nested Gaussian quadrature rules associated with the

distribution of ξ_i. For example the Gauss-Patterson rule is used for uniform variables and Genz-Keister rule for Gaussian variables. For functions of lower regularity Ξ_l are typically chosen to be equidistantly spaced. The number of points m_l of a one-dimensional grid of a given level is dependent on the growth rate of the quadrature rule chosen.

The multivariate basis functions Ψ_k are a tensor product of one dimensional basis functions. Adopting the multi-index notation use above we have

$$\Psi_{\mathbf{l},\mathbf{i}}(\boldsymbol{\xi}) = \prod_{n=1}^{d} \psi_{l_n,i_n}(\xi_n) \tag{4.2}$$

where $\mathbf{i}$ determines the location of a given grid point. There is a one-to-one relationship between Ψ_k in (4.1) and $\Psi_{\mathbf{l},\mathbf{i}}$ and each $\Psi_{\mathbf{l},\mathbf{i}}$ is uniquely associated with a grid point $\boldsymbol{\xi}_{\mathbf{l},\mathbf{i}} = (\xi_{l_1,i_1}, \ldots, \xi_{l_d,i_d}) \in \Xi_{\mathbf{l}}$. Many different one-dimensional basis functions $\psi_{l_n,i_n}(\xi_n)$ can be used. If $J_h(\boldsymbol{\xi})$ has a smooth dependence on $\boldsymbol{\xi}$ then the best choice is the one-dimensional Lagrange polynomials, If local approximation is required one can use a multi-dimensional piecewise polynomial basis [3].

The multi-dimensional basis (4.2) spans the discrete space $V_{\mathbf{l}} \subset L^2(\Gamma_{\boldsymbol{\xi}})$

$$V_{\mathbf{l}} = \text{span}\left\{\Psi_{\mathbf{l},\mathbf{i}} : \mathbf{i} \in \mathcal{K}_{\mathbf{l}}\right\} \quad \mathcal{K}_{\mathbf{l}} = \{\mathbf{i} : i_k = 0, \ldots, m_{l_k}, k = 1, \ldots, d\}$$

These discrete spaces can be further decomposed into hierarchical difference spaces

$$W_{\mathbf{l}} = V_{\mathbf{l}} \setminus V_{\mathbf{l}} \bigoplus_{n=0}^{d} V_{\mathbf{l}-\mathbf{e}_n}$$

The subspaces $W_{\mathbf{l}}$ consists of all basis functions $\Psi_{\mathbf{l},\mathbf{i}} \in V_{\mathbf{l}}$ which are not included in any of the spaces $V_{\mathbf{k}}$ smaller than V_l, i.e. with $\mathbf{k} < \mathbf{l}$. These hierarchical difference spaces can be used to decompose the input space such that

$$V_{\mathbf{l}} = \bigoplus_{\mathbf{k} \le \mathbf{l}} W_{\mathbf{l}} \quad \text{and} \quad L^2(\Gamma_{\boldsymbol{\xi}}) = \bigoplus_{k_1=0}^{\infty} \cdots \bigoplus_{k_d=0}^{\infty} W_{\mathbf{k}} = \bigoplus_{\mathbf{k} \in \mathbb{R}^d} W_{\mathbf{k}}$$

For numerical purposes we must truncate the number of difference spaces used to construct V. Traditional isotropic sparse grids can be obtained by all hierarchical subspaces $W_{\mathbf{l}}$ with and index set that satisfy

$$\mathcal{L} = \{\mathbf{l} : |\mathbf{l}|_1 \le l\} \tag{4.3}$$

Given a truncation, such as the a-priori one above or one which has been determined adaptively, $J_h(\boldsymbol{\xi})$ can be approximated by

$$J_{h,n}(\boldsymbol{\xi}) = \sum_{\mathbf{l}\in\mathcal{L}} J_{\mathbf{l}}, \quad J_{\mathbf{l}} = \sum_{\mathbf{i}\in\mathcal{I}_{\mathbf{l}}} v_{\mathbf{l},\mathbf{i}}\, \Psi_{\mathbf{l},\mathbf{i}}(\boldsymbol{\xi}) \tag{4.4}$$

where $\mathcal{I}_{\mathbf{l}} = \{\mathbf{i} : \Psi_{\mathbf{l},\mathbf{i}} \in W_{\mathbf{l}}\}$.

Here we note that the $v_{\mathbf{i},\mathbf{j}}$ are the coefficient values of the hierarchical product basis, also known as the hierarchical surplus. The surpluses are simply the difference between the function value and the sparse grid approximation at a point, not already in the sparse grid. That is

$$v_{\mathbf{i},\mathbf{j}} = J(\boldsymbol{\xi}_{\mathbf{l},\mathbf{i}}) - J_n(\boldsymbol{\xi}_{\mathbf{l},\mathbf{i}}), \quad \mathcal{L} \cup \mathbf{l} = \emptyset$$

Sparse grids can also be used to interpolate the forward solution $\mathbf{z}_h$ and the adjoint solution $\boldsymbol{\phi}_h$. We define these approximations as $I_n[\mathbf{z}_h] := \mathbf{z}_{h,n}$ and $I_n[\boldsymbol{\phi}_h] := \boldsymbol{\phi}_{h,n}$. When computing a-posteriori error estimates, interpolants of $\mathbf{z}_h$ and $\boldsymbol{\phi}_h$ must be computed at each point in the finite element mesh and step in the time discretiziation.

Although a vast improvement of full tensor grid approximations the number of sparse grid points in an isotropic sparse grid still grows quickly with dimension. To enhance the convergence of sparse grids two types of adaptivity have been developed. The first type of adaptivity refines the grid dimension by dimension [11], greedily choosing points in dimensions that are deemed by the algorithm to be more important. The second type of adaptivity refines the sparse grid locally in regions considered important [14]. When using localized basis functions adaptivity can be performed combining the strengths of both dimension and regional adaptivity [16].

4.1. Dimension adaptivity

The dimension adaptive algorithm begins with a low-level isotropic sparse grid approximation with a set of subspaces $\mathcal{L}$ and active subspaces $\mathcal{A}$. Often $\mathcal{L} = W_{\mathbf{0}}$ and $\mathcal{A} = \{W_{\mathbf{e}_k}, k = 1\ldots, d\}$. We then chooses $W_{\mathbf{l}} \in \mathcal{A}$ with the largest error indicator $\gamma_{\mathbf{l}}$ and refine that subspace. The subspace is refined by adding all indices $W_{\mathbf{k}}$ with $\mathbf{k} = \mathbf{l} + \mathbf{e}_n$, $n = 1, \ldots, d$ that satisfy the following admissibility criterion

$$\mathbf{l} - \mathbf{e}_k \in \mathcal{L} \text{ for } 1 \leq k \leq d,\, l_k > 1 \tag{4.5}$$

Each subspace that satisfies (4.5) is then placed in the active set $\mathcal{A}$. This process continues until a computational budget limiting the number of model

samples (grid points) is reached or a global error indicator drops below a predefined threshold. Pseudo-code for the dimension adaptive algorithm is shown in Algorithm 1. The auxiliary `TERMINATE` routine is given in Algorithm 2.

Purposefully, Algorithm 1 does not specify the contents of the `INDICATOR` and `GLOBAL_INDICATOR` routines in Algorithm 1. These routines control which subspaces are added to the sparse grid. The key to adaptive sparse grids working well is versions of these two routines which respectively provide accurate estimates of the contribution of a subspace to reducing the error in the interpolant, and the error in the entire interpolant.

Typically these indicator functions are functions of the hierarchical surplus values $v_{\mathbf{i},\mathbf{j}}$ of points in the grid. Throughout this paper we will use the indicator

$$\gamma_{\mathbf{l}} = \sum_{\mathbf{i} \in \mathcal{I}_{\mathbf{l}}} |v_{\mathbf{i},\mathbf{j}}| \, w_{\mathbf{i},\mathbf{j}} \tag{4.6}$$

as the baseline for comparison for the proposed method. Here $w_{\mathbf{i},\mathbf{j}}$ is the quadrature weight of the grid point $\xi_{\mathbf{i},\mathbf{j}}$. In addition, we use the global indicator

$$\eta = \sum_{\mathbf{l} \in \mathcal{A}} \gamma_{\mathbf{l}}$$

It is possible alternative definitions of $\gamma_{\mathbf{l}}$ and η could produce more accurate sparse grids for a given number of function evaluations. Alternatives could be generated via studies such as that discussed in [20]. Such an exploration of alternative surplus based criteria, however, is beyond the scope of this paper. Instead this paper focuses on alternative criteria that do not require additional evaluations of the simulation model. These alternative criteria are discussed in Section 6. The differences between surplus refinement and the alternative refinement criteria proposed in Section 6 motivate the conditionals in Algorithm 1 not usually seen in the literature [11, 16].

4.2. Local adaptivity

Locally-adapted sparse grids can be adapted using a similar method to that employed for dimension adapted grids. Instead of refining the grid subspace by subspace, locally-adaptive grids are adapted point by point. When a new point is chosen for refinement it is added to the sparse grid. Children of this point are then found and added to an active set $\mathcal{A}_{\text{local}}$. In this paper we will consider the typical local refinement strategy [17] and the simultaneous dimension and local refinement proposed in [16]. In this manuscript we will refer to the former as traditional local refinement and the

Algorithm 1 `INTERPOLATE`$[f(\boldsymbol{\xi}),\mathcal{L},\mathcal{A},\tau,n]\rightarrow f_n$

For a given $\mathcal{L}$ the points in the sparse grid are $\Xi := \bigcup_{\mathbf{l}\in\mathcal{L}} \Xi_{\mathbf{l}}$.
The number of sparse grid points are $N = \#\Xi$

WHILE NOT `TERMINATE`$[\mathcal{A},N,\tau,n]$

- $W := \arg\max_{W_{\mathbf{l}}\in\mathcal{A}} \gamma_{\mathbf{l}}$ % Determine the subspace with the highest priority
- $\mathcal{A} := \mathcal{A} \setminus W$ % Remove W from the active set
- IF (NOT using surplus refinement) $\mathcal{L} := \mathcal{L} \cup W$
- $\mathcal{J}$:= `REFINE`$[W,\mathcal{L}]$ % Find all admissible forward neighbors of W
- $\gamma_{\mathbf{l}}$:= `INDICATOR`$[W_{\mathbf{l}}]$ $\forall\ W_{\mathbf{l}} \in \mathcal{J}$ % Calculate the priority of the neighbors
- $\mathcal{A} := \mathcal{A} \cup \mathcal{J}$ % Add the forward neighbors to the active index set
- IF (using surplus refinement) $\mathcal{L} := \mathcal{L} \cup \mathcal{J}$

Algorithm 2 `TERMINATE`$[\mathcal{A},N,\tau,n]$

- η := `GLOBAL_INDICATOR`$[\mathcal{A}]$
- IF $\mathcal{A} = \emptyset$ or $N \geq n$ or $\eta < \tau$ RETURN TRUE
- ELSE RETURN FALSE

later as generalised local refinement. Error indicators, are then computed at each point in $\mathcal{A}_{\text{local}}$. The next point chosen for refinement is simply the point $\boldsymbol{\xi}_{\mathbf{l},\mathbf{i}}$ in $\mathcal{A}_{\text{local}}$ with the largest error indicator $\gamma_{\mathbf{l},\mathbf{i}}$. The local adaptive sparse grid procedure can be obtained from Algorithm 1 by defining $\mathcal{A}$ and $\mathcal{L}$ to contain points rather than subspaces and W and $W_{\mathbf{l}}$ to be grid points not subspaces. The `REFINE` routine must also be changed as discussed in [16].

Throughout this paper we will use the indicator

$$\gamma_{\mathbf{l},\mathbf{i}} = |v_{\mathbf{i},\mathbf{j}}|\, w_{\mathbf{i},\mathbf{j}} \tag{4.7}$$

as the baseline for comparison for the proposed method. Here $w_{\mathbf{i},\mathbf{j}}$ is the quadrature weight of the grid point $\xi_{\mathbf{i},\mathbf{j}}$. In addition, we use the global indicator

$$\eta = \sum_{\{\mathbf{l},\mathbf{i}\}\in\mathcal{A}} \gamma_{\mathbf{l},\mathbf{i}}$$

5. Adjoint-based error estimates for samples of a sparse grid surrogate

In many practical situations, we are interested in computing probabilistic quantities, such as the probability of a particular event, from the surrogate approximation which usually requires sampling the surrogate according to the distribution of the random parameters over $\Gamma_{\boldsymbol{\xi}}$. In order to have confidence in our estimates of these probabilistic quantities, we must consider the error in each sample of the surrogate.

In this paper, we employ the technique introduced in [5] and further analyzed in [4, 6] to estimate the error in a sample of the sparse grid surrogate of the quantity of interest.

Given sparse grid approximations of the forward $\mathbf{z}_{h,n}$ and adjoint $\boldsymbol{\phi}_{h,n}$ solutions we can compute the following approximate error estimate

$$J(\boldsymbol{\xi}) - J_{h,n}(\boldsymbol{\xi}) \approx \varepsilon(J_{h,n}(\boldsymbol{\xi})), \tag{5.1}$$

where

$$\begin{aligned}\varepsilon(J_{h,n}(\boldsymbol{\xi})) &= (\boldsymbol{\phi}_{h,n}(x,0,\boldsymbol{\xi}), \mathbf{z}(x,0,\boldsymbol{\xi}) - \mathbf{z}_{h,n}(x,0,\boldsymbol{\xi})) \\ -\int_0^T &\left(\left(\frac{\partial \mathbf{z}_{h,n}(x,t,\boldsymbol{\xi})}{\partial t}, \boldsymbol{\phi}_{h,n}(x,t,\boldsymbol{\xi}) \right) + \boldsymbol{a}(\boldsymbol{\xi}; \mathbf{z}_{h,n}(x,t,\boldsymbol{\xi}), \boldsymbol{\phi}_{h,n}(x,t,\boldsymbol{\xi})) \right) dt.\end{aligned}$$

To construct the approximations of the forward and adjoint solutions we construct a sparse grid approximation for each degree of freedom in the

entire forward and adjoint solutions. Note that unlike the sparse grid approximation of a scalar function given in (4.4), the approximations of $\mathbf{z}_{h,n}$ and $\boldsymbol{\phi}_{h,n}$ are functions, not just of $\boldsymbol{\xi}$, but also the spatial and temporal mesh locations $\mathbf{x}$ and $\mathbf{t}$. However, building approximations over $\mathbf{x}$ and $\mathbf{t}$ requires no additional PDE solves as one solve will produce an entire spatial and temporal history. We simply need to store and compute the hierarchical surplus for each $\mathbf{x}$ and $\mathbf{t}$ separately. This is not as memory intensive as it may first appear. We only need to store the additional hierarchical surpluses at each sparse grid point. The remaining sparse grid data structures do not need to be replicated. We do remark, however, that if the PDE solution has millions or even billions of degrees of freedom then memory problems associated with processing big data sets will arise. This issue is not just endemic to the approach proposed in this paper, but is a difficulty also faced by a posteriori error analysis for deterministic PDEs. The typical mitigation strategy to avoid saving the full forward approximation is called checkpointing [add cites], which effectively balances the cost in saving the solution versus recomputing the solution. Unfortunately, this is not an option for the evaluation of the error estimate (5.1) since neither the forward problem nor the adjoint problem are actually solved at the sample point. We could utilize data compression to reduce the storage costs but this is beyond the scope of this paper and will be pursued in later work.

While the a posteriori error estimate given by (5.1) is only an approximation of the true error, it was shown in [4, 6] that the *error in the error estimate* can be bounded by the product of the pointwise errors (in $\Gamma_{\boldsymbol{\xi}}$) in the forward and adjoint approximation. Thus, for smooth problems the error estimate is higher order than the surrogate approximation and can be used as either a higher order correction term (as in [4, 6]) or to guide stochastic adaptivity as we pursue in Section 6.

6. Alternative adaptivity strategies

Hierarchical surplus indicators such as those in (4.6) and (4.7) require the model to be evaluated at the associated grid point $\boldsymbol{\xi}_{\mathbf{l},\mathbf{i}}$ before the surplus and thus indicator can be computed. This procedure adds a grid point and then checks if that refinement should have been performed. Efficiency can be gained by using an error indicator that flags the need for refinement without the need to evaluate the simulation model. The a posteriori error estimate (5.1) is one such indicator.

We construct the following error indicators using the a posteriori error

estimate for dimension adaptive sparse grids

$$\gamma_{\mathbf{l}}^{\varepsilon} = \sum_{\mathbf{i} \in \mathcal{I}_{\mathbf{l}}} |\varepsilon(J_{h,n}(\boldsymbol{\xi}_{\mathbf{l},\mathbf{i}}))| \, w_{\mathbf{i},\mathbf{j}}, \quad \eta = \sum_{\mathbf{l} \in \mathcal{A}} \gamma_{\mathbf{l}} \tag{6.1}$$

and for locally adaptive sparse grids

$$\gamma_{\mathbf{l},\mathbf{i}}^{\varepsilon} = |\varepsilon(J_{h,n}(\boldsymbol{\xi}_{\mathbf{l},\mathbf{i}}))| \, w_{\mathbf{i},\mathbf{j}}, \quad \eta = \sum_{\{\mathbf{l},\mathbf{i}\} \in \mathcal{A}} \gamma_{\mathbf{l},\mathbf{i}} \tag{6.2}$$

The use of these indicators requires a minor modification to the interpolate algorithms presented in the literature [11, 16]. The modification, present in Algorithm 1, controls when the model is evaluated at a sparse grid point and when that point is added to the sparse grid. For surplus based refinement the model is evaluated at the sparse grid when `INDICATOR` is called on a subspace containing the point (dimension refinement) or on the point directly (local refinement). The point is then immediately added to the sparse grid. When using a posteriori refinement the model does not need to be evaluated until `REFINE` is called on the point at which time the point is also added to the sparse grid.

When using local refinement one should ensure all the ancestors of and point in the sparse grid also exist in the sparse grid. Similarly to when deciding to evaluate the model at a point, the use of a posteriori error guided refinement also effects the point in `INTERPOLATE` that the ancestors are added to the sparse grid. Although the function values at ancestor points are not necessary to compute the error estimate at the child point we add all ancestors to the sparse grid before the error estimate is computed. Alternatively one can also add the ancestor point later, only when the point is added to the grid. However we found during our investigation that adding ancestors before computing the error estimate produced a more accurate error estimate at the candidate points which resulted in a more accurate sparse grid for a fixed number of sample functions for all examples presented in this paper. These results are not shown as the authors believed this minor point would distract from the main conclusions that can be drawn from the results section.

Throughout the remainder of this mauscript we will use the following notation to denote the various types of refinement strategies. Let $\gamma_{\mathbf{l}}$ and $\gamma_{\mathbf{l}}^{\varepsilon}$ respectively denote, hierarchical surplus and aposteriori based dimension based refinement. Let hierarchical surplus based traditional and generalised local refinement be denoted by $\gamma_{\mathbf{l},\mathbf{i}}^{\text{trad}}$ and $\gamma_{\mathbf{l},\mathbf{i}}^{\text{gen}}$. Finally we define $\gamma_{\mathbf{l},\mathbf{i}}^{\varepsilon,\text{gen}}$ to be aposterori error based generalised local refinement.

7. Sparse grid approximations of the error

As previously mentioned, the evaluation of the a posteriori error estimate using (5.1) requires the full forward and adjoint approximations at $\boldsymbol{\xi} \in \Gamma_{\boldsymbol{\xi}}$ as well as the evaluation of the space-time weighted-residual. While this is usually much cheaper than solving the forward problem at $\boldsymbol{\xi}$, the computation cost in producing the error estimate at $\boldsymbol{\xi}$ should not be neglected. If we only need to compute a small number of samples, then this is probably not a significant issue. But we often require a very large number of samples of the surrogate, and consequently a very large number of error estimates, so we propose constructing a surrogate of the error estimate. This can significantly reduce the number of error estimates required and the surrogate of the error estimate can be chosen to have similar accuracy as the point-wise evaluation of (5.1).

Given a sparse grid approximation of the quantity of interest $J_{h,n}(\boldsymbol{\xi})$ we can use a finite sample of error estimates $\varepsilon(J_{h,n}(\boldsymbol{\xi}_{\mathbf{l},\mathbf{i}}))$ at new grid points to build an enhanced sparse grid. Once built the enhanced sparse grid interpolant can be sampled directly during post-processing, thus removing any further need to evaluate (5.1).

The procedure needed to construct an enhanced sparse grid is outlined in Algorithm 3. To build an enhanced sparse grid we must first construct sparse grid approximations of the QOI, forward solution and the adjoint solution. The number of points used in these approximations is controlled by two parameters τ and n. The tolerance τ should be chosen such that refinement is stopped when the interpolation error in the enhanced sparse grid approximation J_n^{ε} becomes less than the *error in the error estimate*. Setting the maximum number of points n to a finite number can also be used as a means to limit the cost of constructing the enhanced sparse grid to within a pre-defined computational budget. The choices of τ and n found in Algorithm 3 are discussed in Section 7.2. Once the approximations of the QOI, forward solution and the adjoint solution have been obtained the sparse grid is enhanced by sampling the error estimate $J_n^{\varepsilon}(\boldsymbol{\xi})$.

An enhanced sparse grid can be built using dimension or local refinement. The construction of the enhanced sparse grid approximation varies slightly depending on whether local or dimension adaptivity is being used. The two types of construction are discussed in Section 7.1.

7.1. Adaptivity

When using dimension-adaptive sparse grids we begin with an initial grid with subspaces in $\mathcal{L}$ and a set of candidate subspaces $\mathcal{A}$ generated

when building the sparse grid approximation of the QOI, forward solution and the adjoint solution. When using surplus refinement $\mathcal{A} \subset \mathcal{L}$ and when using a posteriori error refinement $\mathcal{L} \cap \mathcal{A} = \emptyset$. This difference arises from the fact that surplus refinement requires the model to be evaluated at all points in the candidate subspaces where as a posteriori refinement does not.

Given the sets $\mathcal{A}$ and $\mathcal{L}$ we simply use $\texttt{INTERPOLATE}[J_n^\varepsilon(\boldsymbol{\xi}),\mathcal{L},\mathcal{A},\tau,m]$ to construct the enhanced sparse grid $J_{n,m}^\varepsilon := I_m[J_n^\varepsilon(\boldsymbol{\xi})]$, where $J_n^\varepsilon(\boldsymbol{\xi}) := J_{h,n}(\boldsymbol{\xi}) + \varepsilon(J_{h,n}(\boldsymbol{\xi}))$. Unlike the construction of $J_{h,n}(\boldsymbol{\xi})$ the construction of J_n^ε does not require the evaluation of the forward solution $\mathbf{z}_h(\boldsymbol{\xi}_{\mathbf{l},\mathbf{i}})$ at each new grid point. Rather, evaluating $J_n^\varepsilon(\boldsymbol{\xi})$ only involves evaluating the un-enhanced sparse grid $J_{h,n}(\boldsymbol{\xi}_{\mathbf{l},\mathbf{i}})$ and computing the error estimate $\varepsilon(J_{h,n}(\boldsymbol{\xi}_{\mathbf{l},\mathbf{i}}))$. Aside from noting this minor distinction, constructing $J_{n,m}^\varepsilon$ requires one additional step. Before $\texttt{INTERPOLATE}[J_n^\varepsilon(\boldsymbol{\xi}),\mathcal{L},\mathcal{A},\tau,m]$ is called the function value at all points in $\mathcal{L}$ must be updated to include the physical discretization error estimate $\delta(J_h(\boldsymbol{\xi}))$. That is for all $\boldsymbol{\xi}_{\mathbf{l},\mathbf{i}} \in \mathcal{L}$ set the function value at that point to $J_h(\boldsymbol{\xi}_{\mathbf{l},\mathbf{i}}) + \delta(J_h(\boldsymbol{\xi}_{\mathbf{l},\mathbf{i}}))$. The hierarchical surpluses at each of these points must also be update accordingly. We do this because any new points to the enhanced sparse grid will have function values $J_{h,n}(\boldsymbol{\xi}_{\mathbf{l},\mathbf{i}}) + \varepsilon(J_{h,n}(\boldsymbol{\xi}_{\mathbf{l},\mathbf{i}}))$, where $\varepsilon(J_{h,n}(\boldsymbol{\xi}_{\mathbf{l},\mathbf{i}}))$ includes the deterministic error estimate. We again remark that to compute the physical discretization error estimate, the method used to compute the adjoint solution must be higher-order than the method used to compute the forward solution.

The method used to constructing an enhanced sparse grid using local refinement is similar to that employed when using dimension refinement. Again we must simply define the initial sets $\mathcal{L}$ and $\mathcal{A}$ and then call $\texttt{INTERPOLATE}$ after adjusting the function values and hierarchical surpluses for all $\boldsymbol{\xi}_{\mathbf{l},\mathbf{i}} \in \mathcal{L}$.

7.2. Balancing the stochastic and physical discretization errors

When solving PDEs using techniques such as the finite element method the physical discretization error will be non-zero. This error must be taken into account when determining the time to terminate the various interpolation steps needed to construct an enhanced sparse grid. The error in the quantity of interest $J_{h,n}(\boldsymbol{\xi})$, can be decomposed into two components

$$\|J(\boldsymbol{\xi}) - J_{h,n}(\boldsymbol{\xi})\| \leq \underbrace{\|J(\boldsymbol{\xi}) - J_h(\boldsymbol{\xi})\|}_{I} + \underbrace{\|J_h(\boldsymbol{\xi}) - J_{h,n}(\boldsymbol{\xi})\|}_{II} \tag{7.1}$$

where: I is the finite element *physical discretization* error; and II is the *stochastic approximation* error introduced by approximating the quantity of interest by a sparse grid interpolant.

Algorithm 3 ENHANCED_INTERPOLATE[$J_h(\boldsymbol{\xi})$,$\mathbf{z}_h(\boldsymbol{\xi})$,$\boldsymbol{\phi}_h(\boldsymbol{\xi})$,$\mathcal{L}$,$\mathcal{A}$,$\tau$,$n$]$\rightarrow$ J_n^ε

Define C to be the ratio of the cost of evaluating the error estimate to the combined cost of a forward and adjoint equation evaluation

- $J_h(\boldsymbol{\xi})_{h,n/4}$ = INTERPOLATE[$J_h(\boldsymbol{\xi})$,τ,$n/4$]
- $\mathbf{z}_{h,n/4}(\boldsymbol{\xi},\mathbf{x},\mathbf{t})$ = INTERPOLATE[$\mathbf{z}_h(\boldsymbol{\xi},\mathbf{x},\mathbf{t})$,$\tau$,$n/4$]
- $\boldsymbol{\phi}_{h,n/4}(\boldsymbol{\xi},\mathbf{x},\mathbf{t})$ = INTERPOLATE[$\boldsymbol{\phi}_h(\boldsymbol{\xi},\mathbf{x},\mathbf{t})$,$\tau$,$n/4$]
- $\delta_{\max} = \max_{\boldsymbol{\xi}_{\mathbf{l},\mathbf{i}}\in\mathcal{A}} \delta(J_h(\boldsymbol{\xi}_{\mathbf{l},\mathbf{i}}))$
- $\gamma_{\max} = \max_{\boldsymbol{\xi}_{\mathbf{l},\mathbf{i}}\in\mathcal{A}} \gamma_{\mathbf{l},\mathbf{i}}$
- $\tau = \max(\delta_{\max}, \gamma_{\max}^2)$
- $J_n^\varepsilon(\boldsymbol{\xi})$ = INTERPOLATE[$J_n^\varepsilon(\boldsymbol{\xi}) + \varepsilon(J_{h,n}(\boldsymbol{\xi}))$,$\tau$,$Cn/2$]

% $\mathbf{z}_{h,n/4}$ and $\boldsymbol{\phi}_{h,n/4}$ are needed to compute $\varepsilon(J_{h,n}(\boldsymbol{\xi}))$.

Much of the existing literature focuses on minimizing the stochastic approximation error, however only a few attempts have been made to discuss or account for the combined effect of deterministic and stochastic approximation error. Error bounds for (II) for isotropic sparse grid approximations of elliptic partial differential equations using Clenhsaw-Curtis or Gaussian abscissa are given in [18]. Traditional adjoint-based a posteriori error analysis can be used to control (I). In this paper, we use adjoint-based error estimates to control both (I) and (II).

When quantifying uncertainty of models, deterministic and stochastic discretization error must all be accounted for. It is inefficient to reduce the stochastic error to a level below the error introduced by the deterministic discretization. The methodology presented in this paper provides a practical means of balancing the stochastic and deterministic discretization errors.

When balancing physical and stochastic discretization errors, we are often faced with the fact that the desired accuracy cannot be reached within the computational budget. Therefore, the goal is to reduce the total error as much as possible within the computational budget. For the approach in this paper, we need to consider the following costs: solving the forward problem, solving the adjoint problem, and evaluating the error estimate (5.1). From our experiments we found it to be advantageous to limit the number

of forward and adjoint solves to be 1/2 of the computational budget, and to utilize the remaining effort to evaluate $\varepsilon(J_{h,n}(\boldsymbol{\xi}))$ to build the enhanced sparse grid. In many cases, using half the computational budget for computing $\varepsilon(J_{h,n}(\boldsymbol{\xi}))$ is overkill. This is especially true if the cost in evaluating the error estimate is 2-3 orders of magnitude cheaper than solving the forward or adjoint problems. Consequently, we use an error indicator to terminate the sparse grid earlier if a desired tolerence is met. Note here we have made the reasonable assumption that the cost of one forward solve and one adjoint solve are equal, thus the need choice of $n/4$ when constructing $\mathbf{z}_{h,n/4}$ and $\boldsymbol{\phi}_{h,n/4}$.

The enhanced sparse grid should not be refined so that the error in the approximation is below the deterministic error. So when $J_n^{\varepsilon}(\boldsymbol{\xi})$ is constructed, we set τ to be the maximum of the aproximate pyhiscal discretization error $\delta_{\max}$ and the approximate stochastic error $\gamma_{\max}$ in the enhanced sparse grid. The error in the error estimate is the product of the error in the approximation of the forward solution and the error in the approximation of the adjoint solution. Consequently a reasonable approximation of the potential accuracy of the enhanced sparse grid is the square of the maximum indicator $\gamma_{\mathbf{l},\mathbf{i}}$ of all points in $\mathcal{A}$.

8. Results

In this section, we provide several numerical examples to illustrate the properties and convergence of the proposed methodology.

8.1. Parameterized linear system

Consider the diagonally dominant parameterized linear system

$$\begin{bmatrix} \xi_1 & \xi_2 \\ \xi_3 & \xi_4 \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix} \quad \text{or in matrix notation} \quad \mathbf{Az} = \mathbf{b} \tag{8.1}$$

with $\xi_1, \xi_3 \in [2, 4]$ and $\xi_2, \xi_4 \in [0, 1]$. The adjoint equation is given by

$$\mathbf{A}^T \boldsymbol{\phi} = \boldsymbol{\psi} \tag{8.2}$$

We take our quantity of interest to be the first component of the solution $\mathbf{z}$, so we set $\boldsymbol{\psi} = [1, 0]^T$.

As an initial step to illustrating the benefits of the proposed methodology, we mirror the work outlined in [4] and demonstrate the accuracy gained by enhancing the interpolant of the linear functional $J_{h,n}(\boldsymbol{\xi})$ with samples from the error estimate $\varepsilon(J_{h,n}(\boldsymbol{\xi}))$.

We build a surplus-based dimension-adaptive sparse grid using Clenshaw-Curtis abscissa and compute $\|J_h(\boldsymbol{\xi}) - J_{h,n}(\boldsymbol{\xi})\|_{\ell_2(\Gamma_{\boldsymbol{\xi}})}$ by sampling $\boldsymbol{\xi}$ using a $100,000$ point Latin hypercube. We then compute error-estimates at the same $100,000$ points and calculate $\|J_h(\boldsymbol{\xi}) - (J_{h,n}(\boldsymbol{\xi}) + \varepsilon(J_{h,n}(\boldsymbol{\xi})))\|_{\ell_2(\Gamma_{\boldsymbol{\xi}})}$. Note in all future examples $100,000$ Latin Hypercube samples will be used to compute the discrete ℓ_2 error, unless otherwise stated.

Figure 1 shows the ℓ^2 error in the functional value and the error in the improved function value as the computational cost increases. We define one unit of cost to be the computational cost of one solve of the forward model and assume that the cost of solving the adjoint equation is also equal to one unit. For this example, we ignore the cost of computing the error estimate. We will challenge the last assumption in §8.2. Note that in this figure and

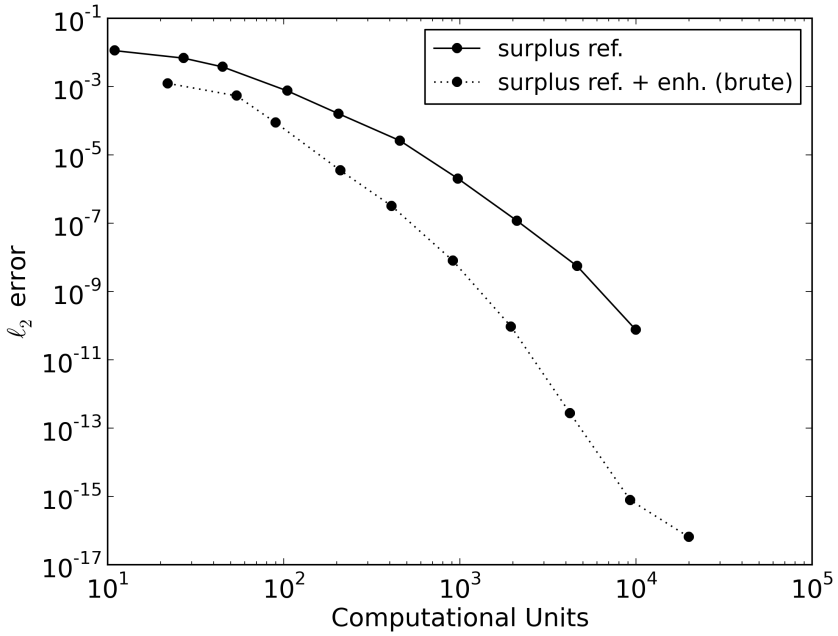


Figure 1: Approximation of the 4d parameterized linear system using a dimension-adaptive Clenshaw-Curtis sparse grid.

all remaining figures, the expression left of the colon in the legend denotes the quantity being approximated and the expression on the right denotes the type of refinement employed. For definitions of the refinement types refer to the end of §6.

Figure 2 compares error in the functional value and the error in the improved function value as the computational cost increases using local adaptivity based upon one-dimensional equidistant meshes and piecewise quadratic basis functions using surplus-based refinement. Throughout the remainder of this manuscript, a quadratic basis will be used when local adaptivity is employed, unless otherwise stated.

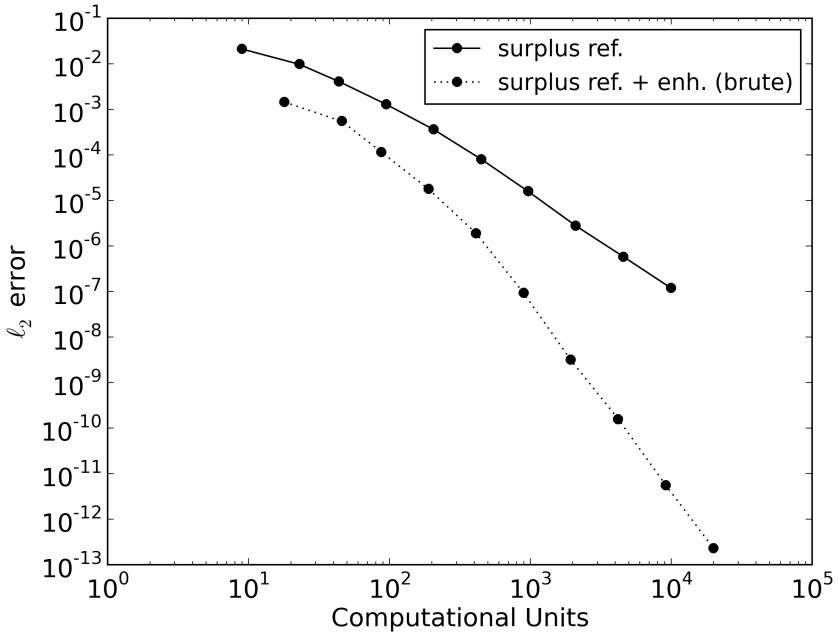


Figure 2: Approximation of the 4d parameterized linear system using a locally-adaptive sparse grid with a quadratic basis

Similarly to the results obtained in [4], which were obtained using enhanced polynomial chaos expansions, enhancing both dimension adaptive and locally adaptive sparse grids with error estimates greatly increases accuracy for a fixed computational cost.

8.2. Heterogeneous diffusion equation

In this section, we investigate the performance of the proposed methodology when applied to a Poisson equation with an uncertain heterogeneous diffusion coefficient. Attention is restricted to the one-dimensional physical space to avoid unnecessary complexity. The procedure described here can easily be extended to higher physical dimensions.

Consider the following problem with $d \geq 1$ random dimensions:

$$-\frac{d}{dx}\left[a(x,\boldsymbol{\xi})\frac{d\mathbf{z}}{dx}(x,\boldsymbol{\xi})\right] = 10, \quad (x,\boldsymbol{\xi}) \in (0,1)\times I_{\boldsymbol{\xi}} \tag{8.3}$$

subject to the physical boundary conditions

$$\mathbf{z}(0,\boldsymbol{\xi}) = 0, \quad \mathbf{z}(1,\boldsymbol{\xi}) = 0 \tag{8.4}$$

Furthermore assume that the log random diffusivity satisfies

$$a(x,\boldsymbol{\xi}) = \bar{a} + \sigma_a \sum_{k=1}^{d} \sqrt{\lambda_k}\phi_k(x)\xi_k \tag{8.5}$$

where $\{\lambda_k\}_{k=1}^d$ and $\{\phi_k(x)\}_{k=1}^d$ are, respectively, the eigenvalues and eigenfunctions of the covariance kernel

$$C_a(x_1,x_2) = \exp\left[-\frac{(x_1-x_2)^2}{2l_c}\right]$$

The variability of the diffusivity field (8.5) is controlled by σ_a and the correlation length l_c which determines the decay of the eigenvalues λ_k. Here we set $d = 25$, $\sigma_a = 1.$, $l_c = 0.1$, $\bar{a} = 0$ and $\xi_k \in [-1,1]$, $k = 1,\ldots,d$ to be independent and uniformly distributed random variables. We choose $d = 25$ so that the error due to truncation of the expansion is below machine precision. We define our quantity of interest to the the value of the solution at x=0.5 which we approximate using the linear functional $J(\boldsymbol{\xi})(\boldsymbol{\xi}) = (\psi(x), \mathbf{z}(x,\boldsymbol{\xi}))$ where

$$\psi(x) = \frac{564.1895835466115}{\pi}\exp(-100(x-.5)^2).$$

The corresponding adjoint problem is given by

$$-\frac{d}{dx}\left[a(x,\boldsymbol{\xi})\frac{d\boldsymbol{\phi}}{dx}(x,\boldsymbol{\xi})\right] = \psi(x), \quad (x,\boldsymbol{\xi}) \in (0,1)\times I_{\boldsymbol{\xi}} \tag{8.6}$$

subject to the physical boundary conditions

$$\boldsymbol{\phi}(0,\boldsymbol{\xi}) = 0, \quad \boldsymbol{\phi}(1,\boldsymbol{\xi}) = 0 \tag{8.7}$$

Unlike the parameterized linear system discussed in the previous section, the solution to (8.3) must be approximated numerically which introduces significant physical discretization error (error labeled I in (7.1)). This physical discretization error must be accounted for when quantifying uncertainty.

We approximate the solution to (8.3) using a standard continuous Galerkin finite element method on a uniform mesh with 100 elements ($h = 0.01$) and piecewise linear basis functions. To include the deterministic component in the error estimate, we need to solve the adjoint problem using a higher-order discretization than was used to solve the forward problem. In this paper, we use a continuous Galerkin method with piecewise quadratic basis functions to approximate the adjoint solution. The deterministic error using the aforementioned discretization is approximately $\delta_{\max} = 8.41 \times 10^{-5}$. Any computational effort used to drive the stochastic error below the deterministic error will be wasted. Figure 3 compares the convergence of an un-enhanced sparse surplus-based, dimension-adaptive, Clenshaw-Curtis grid $J_{h,n}(\boldsymbol{\xi})$ with the enhanced sparse grid approximation J_n^{ε}. Again we have computed the error estimates directly at the $100,000$ point Latin hypercube samples used to compute $\|J_h(\boldsymbol{\xi}) - J_{h,n}(\boldsymbol{\xi})\|_{\ell_2(\Gamma_{\boldsymbol{\xi}})}$ and $\|J_h(\boldsymbol{\xi}) - (J_{h,n}(\boldsymbol{\xi}) + \varepsilon(J_{h,n}(\boldsymbol{\xi})))\|_{\ell_2(\Gamma_{\boldsymbol{\xi}})}$. Clearly, again there is significant benefit from using error estimates to improve the sparse grid interpolant.

Figure 3 also shows the importance of not refining the sparse grid when the stochastic error is smaller than the physical discretization error. When this occurs the error estimate which includes the physical discretization error the accuracy in the enhanced sparse grid stagnates at approximately the physical discretization error.

The cost of computing $\varepsilon(J_{h,n}(\boldsymbol{\xi}))$ is non-trivial and in this example the cost of evaluating $\varepsilon(J_{h,n}(\boldsymbol{\xi}))$ at $100,000$ points would far outweigh the cost of the forward and adjoint solves. Consequently we must build an approximation of the error estimate using the approach described in Section 7. The need to set appropriate termination criteria outlined in Section 7.2 is illustrated in Figure 4. The figure illustrates that as more forward and adjoint solutions are used to construct $J_{h,n}(\boldsymbol{\xi})$ the higher accuracy that can be obtained in the enhanced sparse grid $J_{n,m}^{\varepsilon}$. However the accuracy that can be obtain is limited, yet any error indicator that does not know the 'true' values of the quantity of interest will appear to converge to machine precision. Consequently, as stated in Section 7.2, we set τ when constructing the enhanced sparse grid in Algorithm 3 to be maximum of the approximate physical discretization error $\delta_{\max}$ and the approximate stochastic error $\gamma_{\max}$ in the enhanced sparse grid.

Figure 5 plots the convergence of the un-enhanced surplus-based dimension-adaptive Clenshaw-Curtis sparse grids $J_{h,n}(\boldsymbol{\xi})$, J_n^{ε} and $J_{n,m}^{\varepsilon}$. Both the un-enhanced sparse grid and the enhanced sparse grid are refined until the estimate of the stochastic error $\gamma_{\max}$ is below the deterministic error δ_{max}.

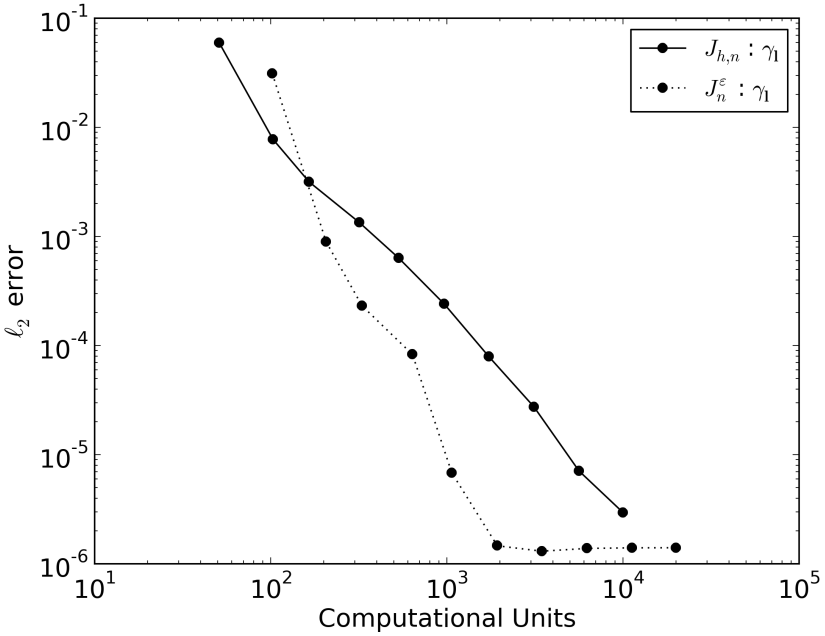


Figure 3: Approximation of the 25d heat equation using a dimension-adaptive Clenshaw-Curtis sparse grid

In the case of the un-enhanced grid this point is reached, and the convergence study is terminated, when the sparse grid has $10,000$ points. The computational cost of $J^\varepsilon_{n,m}$ includes the cost of evaluating the residual to compute the error estimates needed at the sparse grid points. Here, we assume that the cost of an error estimate is $1/25$ of the cost of one forward solve.

The enhanced sparse grid $J^\varepsilon_{n,m}$ is significantly more accurate than the un-enhanced grid. However $J^\varepsilon_{n,m}$ is not as accurate as J^ε_n. This is not a fair comparison as in practice, using J^ε_n is typically computationally infeasible. However the comparison does that the choice of termination conditions in Algorithm 1 means that the grid terminates before the error in the enhanced grid is below the error in the error estimate. As Figure 4 shows this level of accuracy can be obtained however it comes at the cost of (possibly many) additional residual computations. It is plausible that a 'better' indicator γ_1 could be constructed and a 'better' chosen however the form of such quantities are problem dependent and beyond the scope of this paper.

Figure 6 plots the convergence of the un-enhanced surplus-based locally-

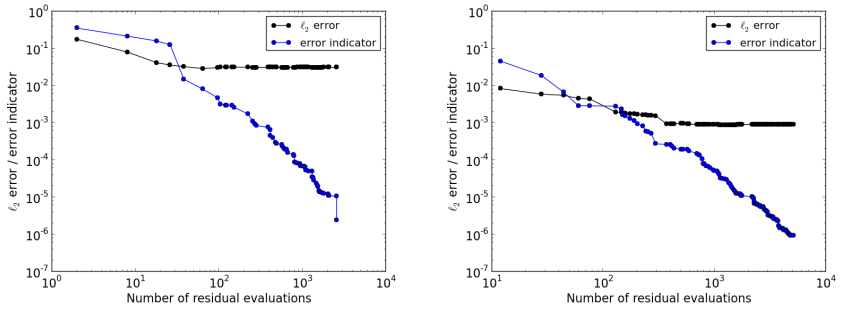


Figure 4: The decay of the ℓ_2 error vs the decay in the error indicator for the enhanced sparse grid $J^{\varepsilon}_{n,m}$, starting from an initial grid with 51 points (left) and 103 points (right).

adaptive sparse grids $J_{h,n}(\boldsymbol{\xi})$, J^{ε}_{n} and $J^{\varepsilon}_{n,m}$. The same conclusions drawn when using dimension adaptivity can also be made here. The improvement obtained by using $J^{\varepsilon}_{n,m}$ instead of $J_{h,n}(\boldsymbol{\xi})$ is reduced however this can be addressed by the use of different refinement strategies discussed in Section 8.4

8.3. Non-linear coupled system of ODEs

For our third model problem, we consider a non-linear system of ordinary differential equations governing a competitive Lotka–Volterra model of the population dynamics of species competing for some common resource. The model is given by

$$\begin{cases} \frac{d\mathbf{z}_i}{dt} = r_i \mathbf{z}_i \left(1 - \sum_{j=1}^{3} \alpha_{ij} \mathbf{z}_j\right), & t \in (0, 10], \\ \mathbf{z}_i(0) = \mathbf{z}_{i,0} \end{cases}, \tag{8.8}$$

for $i = 1, 2, 3$. The initial condition, $\mathbf{z}_{i,0}$, and the self-interacting terms, α_{ii}, are given, but the remaining interaction parameters, α_{ij} with $i \neq j$ as well as the re-productivity parameters, r_i, are unknown. We assume that these parameters are each uniformly distributed on $[0.3, 0.7]$. We approximate the solution to (8.8) in time using a Backward Euler method with 1000 time steps ($\Delta t = 0.01$) which gives a deterministic error of approximately 1.00×10^{-4}.

The quantity of interest is the population of the third species at the final

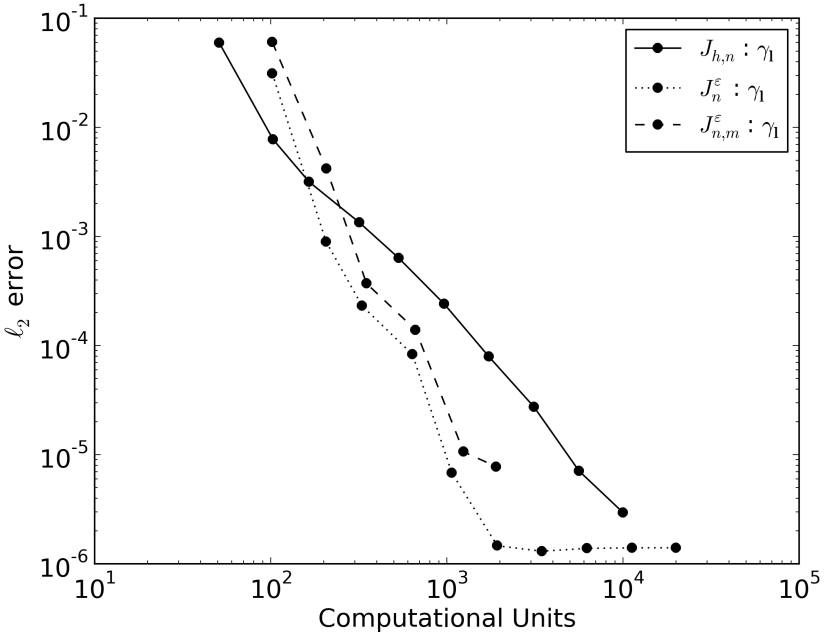


Figure 5: Comparison of the ℓ_2 convergence of the enhanced dimension-adaptive sparse grid using direct evaluation of the error estimate J_n^ε and a sparse grid approximation of the enhanced QOI $J_{n,m}^\varepsilon$.

time, $\mathbf{z}_3(10)$. The corresponding adjoint problem is

$$\begin{cases} -\frac{d\boldsymbol{\phi}_i}{dt} = r_i \boldsymbol{\phi}_i \left(1 - \sum_{j=1}^3 \alpha_{ij} \mathbf{z}_j\right) + r_i \mathbf{z}_i \left(1 - \sum_{j=1}^3 \alpha_{ji} \boldsymbol{\phi}_j\right), & t \in (10, 0], \\ \boldsymbol{\phi}_i(10) = 0, & i = 1, 2 \\ \boldsymbol{\phi}_3(10) = 1. \end{cases}, \tag{8.9}$$

We approximate the adjoint solution in time using a second-order Crank-Nicholson method with the same number of time steps.

Figures 7 and Figures 8 compares the convergence of $J_{h,n}(\boldsymbol{\xi})$, J_n^ε and $J_{n,m}^\varepsilon$ when using a surplus based dimension-adaptive Clenshaw-Curtis sparse grid and surplus locally adaptive sparse grid, respectively. Again, significant increases in accuracy can be achieved by using $J_{n,m}^\varepsilon$ instead of $J_{h,n}(\boldsymbol{\xi})$. And as seen before the the accuracy of $J_{n,m}^\varepsilon$ is not as high as J_n^ε but could be improved at the cost of additional residual calculations (error estimates) at new sparse grid points.

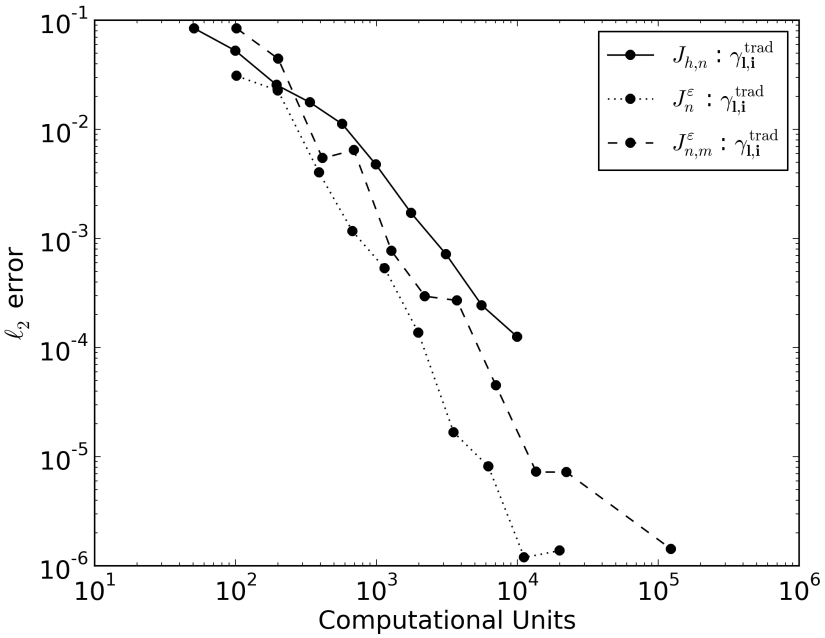


Figure 6: Comparison of the ℓ_2 convergence of the enhanced locally-adaptive sparse grid using direct evaluation of the error estimate and a sparse grid approximation of the enhanced QOI

8.4. Alternative Refinement Strategies

In this section we demonstrate the utility of using the alternative refinement strategies. outlined in Section 6. Specifically we consider a posteriori refinement indicators for both dimension and locally adapted sparse grid. In the case of local refinement we also consider the simultaneous dimension and local refinement in addition to the typical local refinement strategy used in all examples thus far.

Figure 9 compares refinement strategies for the 25d heat equation problem presented in Section 8.2. It is clear that the use of a posteriori refinement results in significant increases in efficiency. This result is due to the fact that the a posteriori subspace refinement indicator $\gamma_{\mathbf{l}}$ only requires computation of error estimates which are relatively cheap compared to the evaluation of the forward model. Thus the function evaluations that are needed by surplus refinement that probe candidate subspaces are often redundant and this redundant evaluation can be avoided by using a posteriori refinement. The high-dimensional nature and the strong degree of anisotropy of this

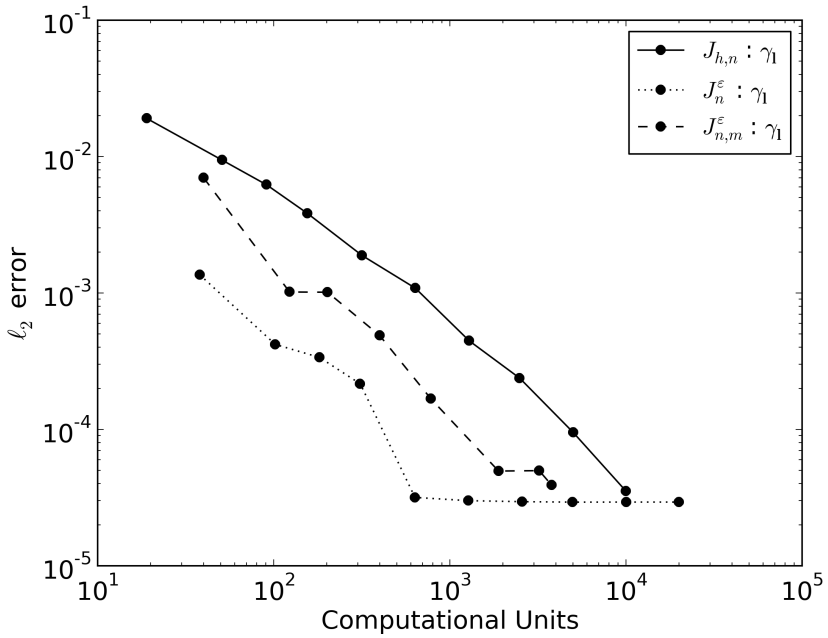


Figure 7: Comparison of the ℓ_2 convergence of the enhanced dimension-adaptive sparse grid using direct evaluation of the error estimate and a sparse grid approximation of the enhanced QOI

problem are an ideal setting for a posteriori refinement. If the function was lower dimensional or have less anisotropy then surplus refinement would be more competitive as the amount of redundant model evaluations would reduce. This point is illustrated in Figure 10 which compares a posteriori and surplus dimension adaptive sparse grids when used to solve the 9d predator-prey model presented in Section 8.3.

A posteriori refinement can also be used in conjunction with local refinement. Figure 11 compares refinement strategies for the 25d heat equation. The use of generalized local refinement results in a vast improvement over traditional local refinement. The best refinement strategy for this problem is to use a posteriori refinement indicators with the generalized local refinement. This result is due to the high-dimension of the problem and the high degree of anisotropy.

Figure 11 compares refinement strategies for the 9d predator prey model. Due to the low degree of anisotropy of this function generalized local refinement provides now improvement over traditional local refinement. In the

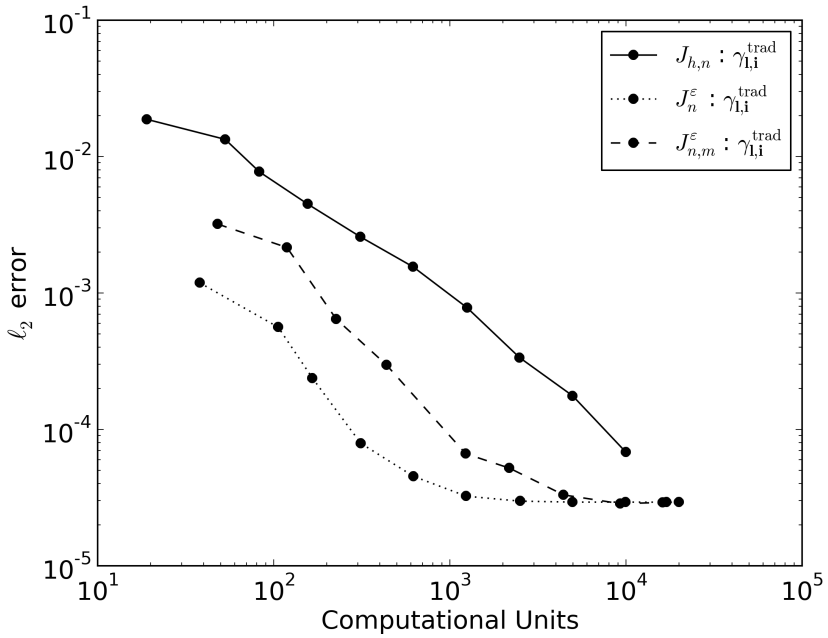


Figure 8: Comparison of the ℓ_2 convergence of the enhanced locally-adaptive sparse grid using direct evaluation of the error estimate and a sparse grid approximation of the enhanced QOI

case of the surplus-based enhanced sparse grid the accuracy is slightly worse. Traditional surplus based refinement is the most accurate strategy here, but the a posteriori error based generalized local refinement has comparable accuracy.

9. Conclusions

In this paper we present an algorithm for adaptive sparse grid approximations of quantities of interest computed from discretized partial differential equations. We use adjoint-based a posteriori error estimates of the interpolation error in the sparse grid to enhance the sparse grid approximation and to drive adaptivity. Using a number of numerical examples we show that utilizing aposteriori error estimates provides significantly more accurate functional values for random samples of the sparse grid approximation. The cost of computing these error estimates can be non-trivial and thus we provide a practical method for enhancing a sparse grid approximation using

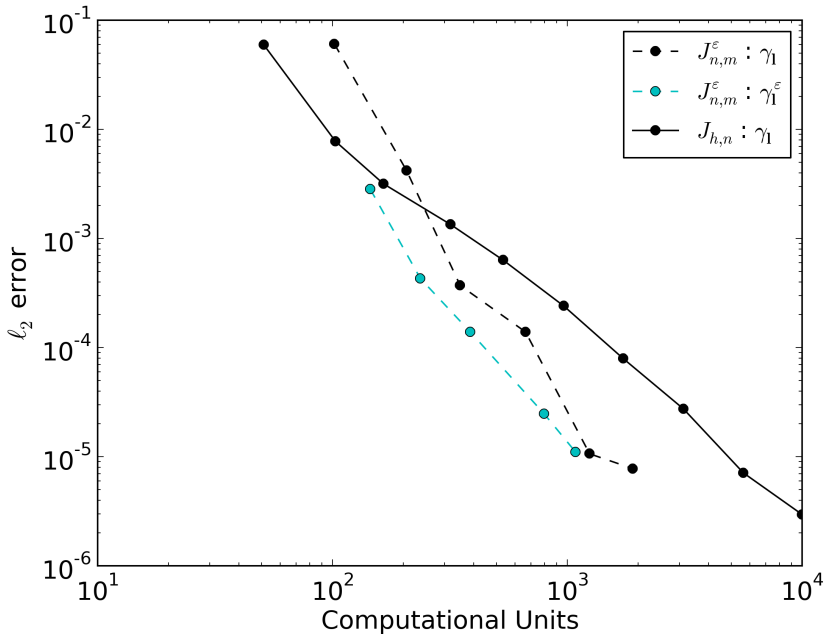


Figure 9: Comparison of a posteriori-error and traditional surplus guided refinement for the 25d heat equation

only a finite set of error estimates.

Aside from using aposteriori error estimates to enhance an approximation we also demonstrate that error estimates can be used to increase the efficiency of adaptive sampling. Traditional sparse grid adaptivity employs error indicators based upon the hierarchical surplus are used to flag dimensions or local regions for refinement. However such approaches require the model to be evaluated at a new point before one can determine if refinement should have taken place. In this paper we numerically demonstrated that refinement using aposteriori error estimates can significantly reduce the amount of redundant sampling compared when compared to traditional hierarchical refinement.

Numerically solving a PDE requires a discretization of the physical domain which introduces a physical discretization error which should be accounted for. In combination with the aforementioned enhancement and refinement procedures we use a posteriori error estimates to ensure that the sparse grid is not refined beyond the point at which the stochastic interpolation error is below the physical discretization error. The methodology

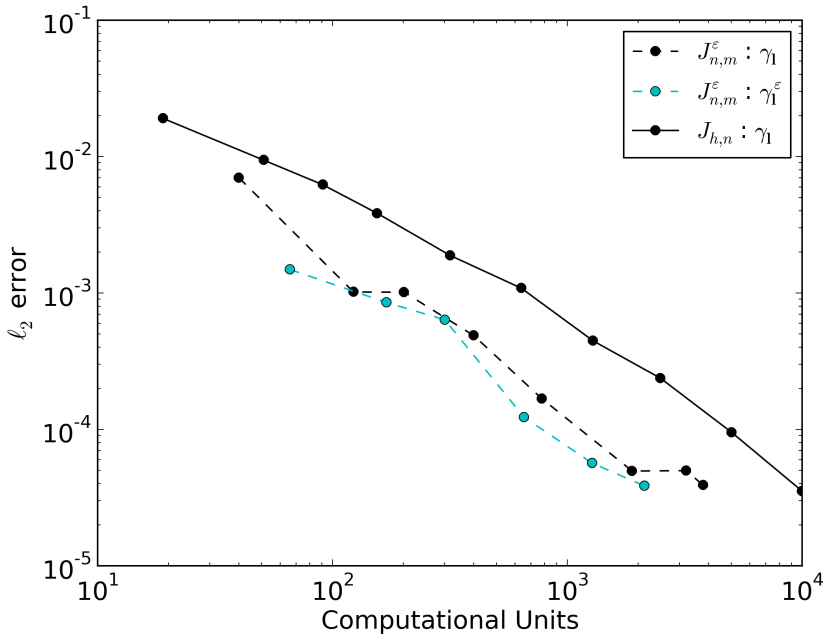


Figure 10: Comparison of a posteriori-error and traditional surplus guided refinement for the 9d predator prey model

presented provides a practical means of balancing the stochastic and deterministic discretization errors.

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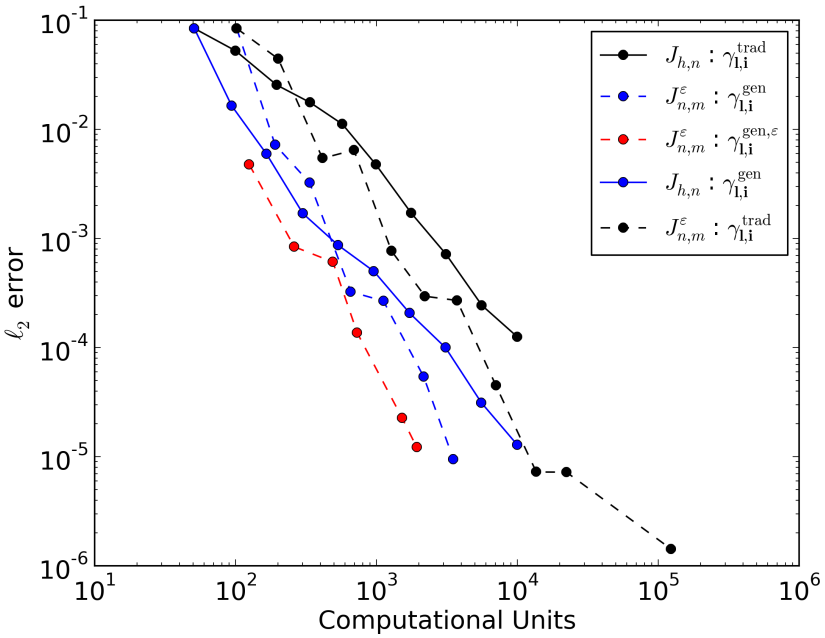


Figure 11: Surplus vs Generalized Surplus heat equation

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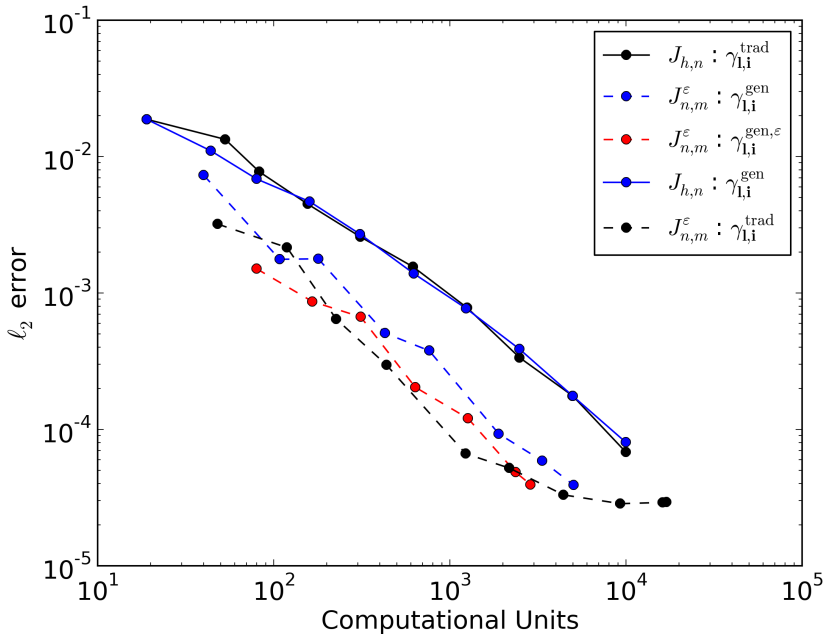


Figure 12: Error vs Generalized error heat equation

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